

Algebra: Applying the Zero Product Property

Practice Worksheet • numberbender.com



Name: _____

Date: _____

Score: ____ / 10

DIRECTIONS

Solve each equation using the Zero Product Property: if $a \cdot b = 0$, then $a = 0$ or $b = 0$. If the equation is not yet factored, factor it first, then set each factor equal to zero.

1. Solve:

$$(x - 3)(x + 5) = 0$$

Answer: _____

2. Solve:

$$x(x - 7) = 0$$

Answer: _____

3. Solve:

$$(2x - 1)(x + 4) = 0$$

Answer: _____

4. Solve:

$$(3x + 6)(x - 2) = 0$$

Answer: _____

5. Solve:

$$x^2 - 9 = 0$$

Answer: _____

6. Solve:

$$x^2 - 5x = 0$$

Answer: _____

7. Solve:

$$x^2 + 7x + 10 = 0$$

Answer: _____

8. Solve:

$$x^2 - x - 12 = 0$$

Answer: _____

9. Solve:

$$2x^2 + 5x - 3 = 0$$

Answer: _____

10. Solve:

$$x^2 + 6x + 9 = 0$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Zero Product Property Worksheet

More free worksheets, video lessons & courses:
numberbender.com/math-worksheets

Page 1 of 2 · Video: youtu.be/BuOrHLgdKas



Watch the lesson



More free sheets

Answer Key & Solutions

Algebra: Applying the Zero Product Property • Numberbender

TEACHER NOTES

Items 1–4: already factored — set each factor to zero. Items 5–6: factor out a common factor or use difference of squares. Items 7–9: factor a trinomial first. Item 10: perfect-square trinomial — double root.

1. Solve: $(x - 3)(x + 5) = 0$

Answer: $x = 3, -5$

$x - 3 = 0 \rightarrow x = 3$; $x + 5 = 0 \rightarrow x = -5$.

2. Solve: $x(x - 7) = 0$

Answer: $x = 0, 7$

$x = 0$ or $x - 7 = 0 \rightarrow x = 7$.

3. Solve: $(2x - 1)(x + 4) = 0$

Answer: $x = 1/2, -4$

$2x - 1 = 0 \rightarrow x = 1/2$; $x + 4 = 0 \rightarrow x = -4$.

4. Solve: $(3x + 6)(x - 2) = 0$

Answer: $x = -2, 2$

$3x + 6 = 0 \rightarrow x = -2$; $x - 2 = 0 \rightarrow x = 2$.

5. Solve: $x^2 - 9 = 0$

Answer: $x = \pm 3$

Difference of squares: $(x - 3)(x + 3) = 0 \rightarrow x = \pm 3$.

6. Solve: $x^2 - 5x = 0$

Answer: $x = 0, 5$

Factor x : $x(x - 5) = 0 \rightarrow x = 0$ or $x = 5$.

7. Solve: $x^2 + 7x + 10 = 0$

Answer: $x = -2, -5$

$(x + 2)(x + 5) = 0 \rightarrow x = -2$ or $x = -5$.

8. Solve: $x^2 - x - 12 = 0$

Answer: $x = 4, -3$

$(x - 4)(x + 3) = 0 \rightarrow x = 4$ or $x = -3$.

9. Solve: $2x^2 + 5x - 3 = 0$

Answer: $x = 1/2, -3$

$(2x - 1)(x + 3) = 0 \rightarrow x = 1/2$ or $x = -3$.

10. Solve: $x^2 + 6x + 9 = 0$

Answer: $x = -3$ (double root)

$(x + 3)^2 = 0 \rightarrow x = -3$.

